

11.11 Classwork: Regression and Normal Distribution (calculator) — Markscheme

1. [Maximum mark: 7]

A sports scientist is studying the relationship between the age of an athlete, a years, and their reaction time, r milliseconds. The following table shows the data collected from eight athletes.

a	18	21	25	28	33	36	40	45
r	155	159	176	178	198	199	215	226

- (a) Find the regression line of r on a in the form $r = ma + c$, where m and c are constants. Give m and c correct to three significant figures. [3]

GDC: Enter data into lists, run linear regression.

$$m = 2.70 \quad (\mathbf{A1})$$

$$c = 105 \quad (\mathbf{A1})$$

$$r = 2.70a + 105 \quad (\mathbf{A1})$$

*Award **(A1)** for each of m and c correct to 3 s.f., and **(A1)** for writing the equation in the required form. Accept $m = 2.70$ (3 s.f.) and $c = 105$ (3 s.f.).*

- (b) Write down the value of Pearson's product-moment correlation coefficient, r . [1]

$$r = 0.994 \quad (\mathbf{A1})$$

Award for $r = 0.994$ (3 s.f.). Full GDC display acceptable as working.

- (c) Use the regression line to estimate the reaction time of an athlete who is 30 years old. [2]

$$r = 2.70(30) + 105 \quad (\mathbf{M1})$$

$$= 186 \text{ ms} \quad (\mathbf{A1})$$

*Award **(M1)** for substituting $a = 30$ into their regression equation. **(A1)** for 186 (3 s.f.). Follow through from their m and c .*

- (d) State whether this estimate is an example of interpolation or extrapolation. Give a reason for your answer. [1]

Interpolation, because 30 is within the range of the data ($18 \leq a \leq 45$). **(R1)**

*Award **(R1)** for "interpolation" with a correct reason referencing the data range. Do not award if the reason is missing or incorrect.*

2. [Maximum mark: 8]

The time, T minutes, that customers wait in line at a coffee shop is normally distributed with a mean of 4.5 minutes and a standard deviation of 1.2 minutes.

(a) Find $P(T > 6)$. [2]

GDC: normalcdf with lower = 6, upper = ∞ , $\mu = 4.5$, $\sigma = 1.2$.

$$P(T > 6) = 1 - P(T \leq 6) \quad (\mathbf{M1})$$

$$= 0.106 \quad (\mathbf{A1})$$

*Award **(M1)** for a correct probability statement or GDC setup. **(A1)** for 0.106 (3 s.f.).*

(b) Find $P(3 < T < 6)$. [2]

GDC: normalcdf with lower = 3, upper = 6, $\mu = 4.5$, $\sigma = 1.2$.

$$P(3 < T < 6) = P(T < 6) - P(T < 3) \quad (\mathbf{M1})$$

$$= 0.789 \quad (\mathbf{A1})$$

*Award **(M1)** for a correct approach (difference of two probabilities or direct GDC entry). **(A1)** for 0.789 (3 s.f.).*

(c) On a particular morning, 10 customers each wait in line independently. Find the probability that exactly 2 of them wait more than 6 minutes. [2]

$$\text{Let } X \sim B(10, 0.1056 \dots) \quad (\mathbf{M1})$$

GDC: binompdf(10, 0.1056, 2)

$$P(X = 2) = 0.206 \quad (\mathbf{A1})$$

*Award **(M1)** for recognizing a binomial distribution with $n = 10$ and p from part (a). Award **(A1)** for a correct answer. Follow through from their answer to part (a). **(A1)** for 0.206 (3 s.f.).*

(d) A customer is selected at random. Given that they waited more than 3 minutes, find the probability that they waited more than 6 minutes. [2]

$$P(T > 6 \mid T > 3) = \frac{P(T > 6)}{P(T > 3)} \quad (\mathbf{M1})$$

$$P(T > 3) = 0.894$$

$$= \frac{0.1056 \dots}{0.8944 \dots} = 0.118 \quad (\mathbf{A1})$$

*Award **(M1)** for correct use of the conditional probability formula. Award **(A1)** for a correct answer. **(A1)** for 0.118 (3 s.f.).*